

HONOURS MEET

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What is Behavioural Cloning ?

1. BC is a supervised learning algorithm for imitation learning where we collect expert demonstrations which lead to expert actions. We train a policy in BC to directly learn $\pi(a | s) \approx \text{Expert}(s)$
2. In our case inputs are perception-based features and outputs are robot actions(linear velocity and angular velocity)
3. BC learns only from the states present in the expert dataset. It learns what the expert did but does not explicitly learn why the behaviour is desirable.
4. Instead of directly copying the expert behaviour Adversarial imitation learning methods learn to distinguish expert behaviour vs policy behaviour

What is GAIL?

1. GAIL -Generative Adversarial Imitation Learning uses two components - the policy and the discriminator. For our project we tried both PPO and SAC. PPO gave out better results
2. The feature extractor contains 16 features per frame i.e. two consecutive frames X 16 features - Goal features like goal distance, goal angle , goal visibility and Obstacle features being 5 proximity/distance features, 5 obstacle angle features and obstacle density and Optical flow features (v_x and v_y)
3. Expert observations + expert actions are collected from the expert controller and each observation is represented using our 32-dim perception feature vector
4. The 32-D feature vector is given to the PPO policy and PPO outputs the robot action: $a = (v, \omega)$. The policy generates its own trajectories and at every time- step we obtain the (s,a) .
5. The Discriminator receives the expert and policy state-action pairs and it learns to distinguish expert behaviour from the policy behaviour

6. The discriminator's output is converted into a reward for the policy.
7. If the policy behaviour looks more like the expert → higher reward. If it looks unlike the expert → lower reward.
8. PPO uses this imitation reward to update the policy. The updated policy generates new trajectories.
9. New policy trajectories are again compared with expert trajectories. The discriminator and policy are repeatedly updated. Eventually, the policy learns to produce expert-like navigation behaviour.

What is AIRL ?

1. Adversarial Inverse Reinforcement Learning also uses Expert demonstrations, policy, discriminator and PPO. But unlike GAIL, AIRL attempts to learn an underlying reward function from expert behaviour.
2. The learned reward is based on $f(s,a,s')=g(s,a)+\gamma h(s')-h(s)$ where
 - $g(s,a) \rightarrow$ learned task-related reward
 - $h(s) \rightarrow$ potential/value-related component
 - $\gamma \rightarrow$ discount factor
3. Like GAIL, the expert trajectories are collected and the observations are given to the PPO policy and PPO produces $a = (v,w)$. The policy interacts with the environment, the action is executed and the policy generates (s,a,s')
4. AIRL discriminator receives expert and policy transitions and it tries to distinguish expert transition and policy transition.
5. AIRL models its rewards function and this attempts to capture the underlying reward structure that explains the expert's behaviour.

6. The learned AIRL reward is provided to PPO and PPO updates its policy to **maximize this learned reward**.

7. The improved policy interacts with the environment again. New (s,a,s') transitions are generated. The discriminator and policy continue to be updated.

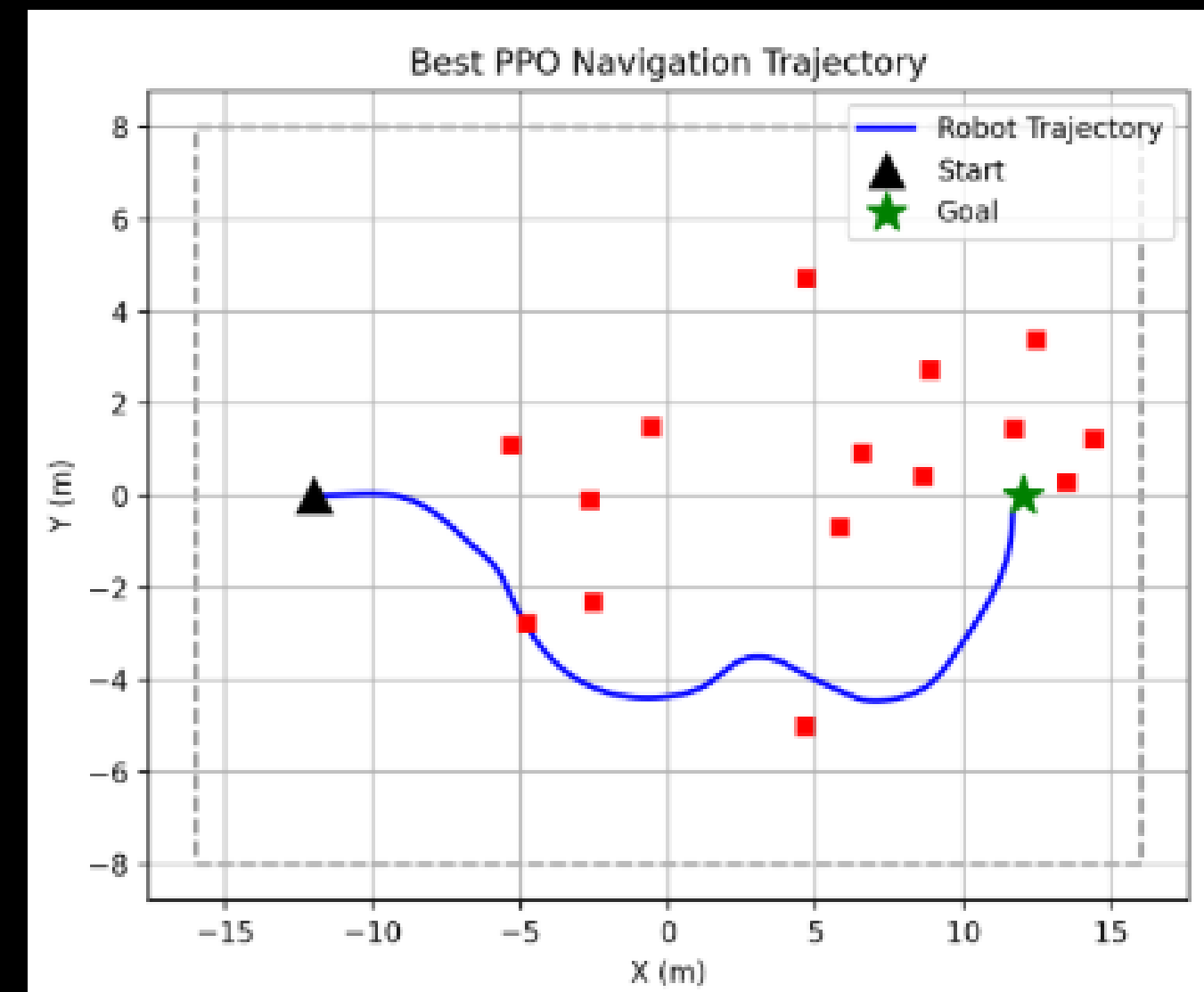
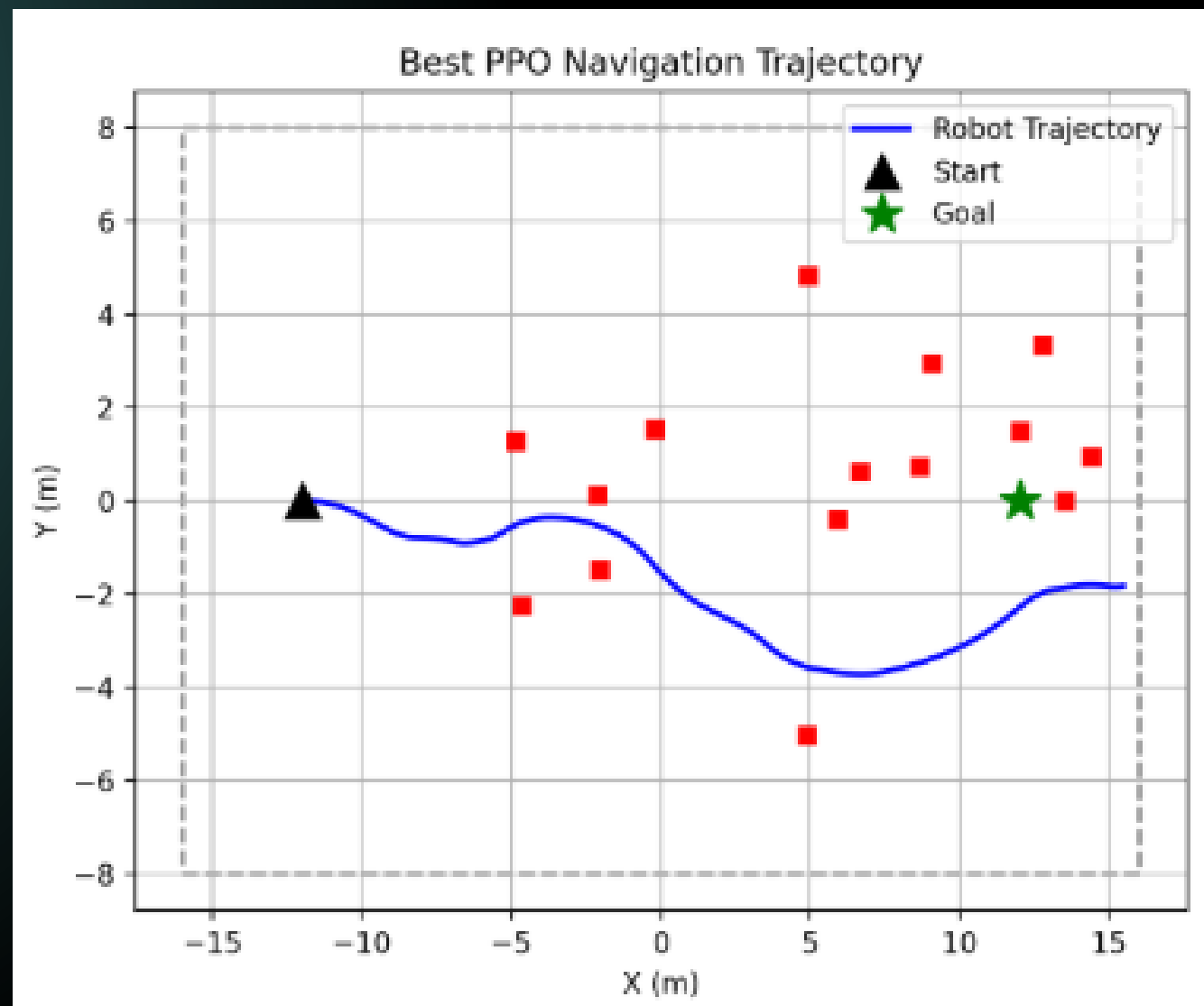
8. AIRL produced poor navigation:

- Robot often moved away from useful goal directions.
- Robot sometimes travelled toward areas where the goal was not visible.
- Frequent wall/obstacle collisions.
- Short trajectories followed by collision.

9. AIRL must learn a meaningful reward structure from the demonstrations.

10. With limited/ambiguous observations, this is significantly harder.

10. Goal information was not explicit enough as the original info heavily relied on goal visibility, goal distance and goal angle. When the goal was not visible, the observation could become ambiguous. AIRL could therefore learn a reward landscape that did not reliably indicate which direction leads toward the goal.
11. The environment contained strong collision/wall penalties. These signals could conflict with each other and PPO could end up optimizing the wrong signal



Experimental Conclusion

1. Learning Objective - Gail matches expert behaviour while AIRL recovers underlying reward
2. Training Complexity in GAIL is lower than AIRL training complexity
3. Dependence on transition quality is lower in GAIL whereas it is higher in AIRL
4. GAIL is consistent with the goal-directed behaviour
5. Collision avoidance is better in GAIL than in AIRL.

Thank You